

GATE 2025 15th Feb 25 S2

Candidate ID	EC25S67432084
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Test Center Name	MUTHAYAMMAL COLLEGE OF ARTS AND SCIENCE
Test Date	15/02/2025
Test Time	2:30 PM - 5:30 PM
Subject	EC ELECTRONICS AND COMMUNICATION ENGINEERING

Section : General Aptitude

- Q.1** The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	Gold	Silver	Bronze
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

Options A.

USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

B.

The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

C. Canada has won twice as many total medals as Japan.

D.

France will occupy the third place if the list were made on the basis of the total number of medals won.

Question Type : MCQ

Question ID : 1422761650

Status : Answered

Chosen Option : A

- Q.2 A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

Note: All lengths are in meter. The figure shown is representative.



- Options
- A. $b_1 < 0.5$
 - B. $b_2 > 0.5$
 - C. $b_2 < b_1 + 0.5$
 - D. $b_1 + b_2 < 1$

Question Type : MCQ

Question ID : 1422761648

Status : Not Attempted and
Marked For Review

Chosen Option : --

- Q.3 Identify the option that has the most appropriate sequence such that a coherent paragraph is formed:

- P. Over time, such adaptations lead to significant evolutionary changes with the potential to shape the development of new species.
- Q. In natural world, organisms constantly adapt to their environments in response to challenges and opportunities.
- R. This process of adaptation is driven by the principle of natural selection, where favorable traits increase an organism's chances of survival and reproduction.
- S. As environments change, organisms that can adapt their behavior, structure and physiology to such changes are more likely to survive.

- Options
- A. $Q \rightarrow S \rightarrow R \rightarrow P$
 - B. $R \rightarrow S \rightarrow Q \rightarrow P$
 - C. $P \rightarrow Q \rightarrow R \rightarrow S$
 - D. $S \rightarrow P \rightarrow R \rightarrow Q$

Question Type : MCQ

Question ID : 1422761647

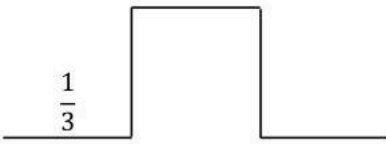
Status : Answered

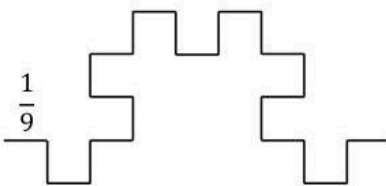
Chosen Option : A

Q.4 The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n ' is:

Note: The figures shown are representative.

Iteration 0: 

Iteration 1:  Length of each segment: $\frac{1}{3}$

Iteration 2:  Length of each segment: $\frac{1}{9}$

Options

A. $\left(\frac{5}{3}\right)^{2n}$

B. $\left(\frac{5}{3}\right)^{\frac{n}{2}}$

C. $\left(\frac{5}{3}\right)^n$

D. $\left(\frac{5}{3}\right)^{n(2n-1)}$

Question Type : MCQ

Question ID : 1422761645

Status : Not Attempted and
Marked For Review

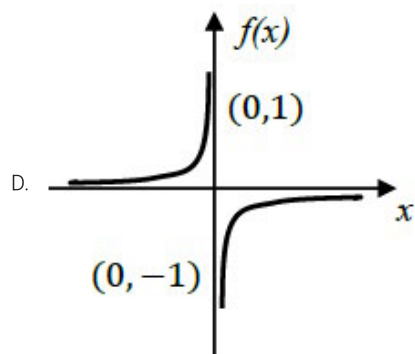
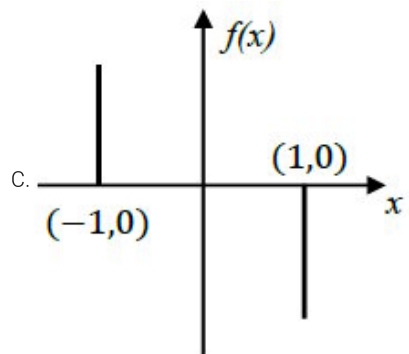
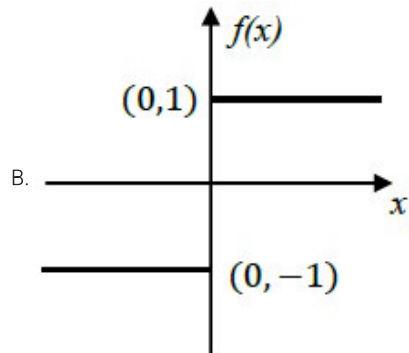
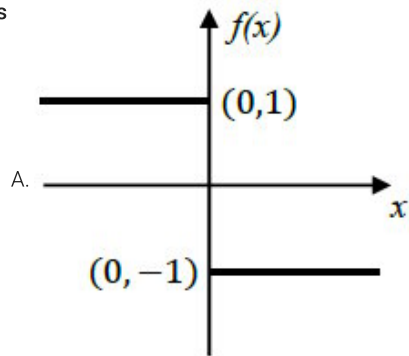
Chosen Option : --

Q.5

Which one of the following plots represents $f(x) = -\frac{|x|}{x}$, where x is a non-zero real number?

Note: The figures shown are representative.

Options



Question Type : MCQ

Question ID : 1422761646

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.6 The 12 musical notes are given as $C, C^\#, D, D^\#, E, F, F^\#, G, G^\#, A, A^\#,$ and B . Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^\#$ and C is:

Options

A. $\sqrt{2}$

B. $\sqrt[4]{2}$

C. 2

D. $\sqrt[6]{2}$

Question Type : MCQ

Question ID : 1422761644

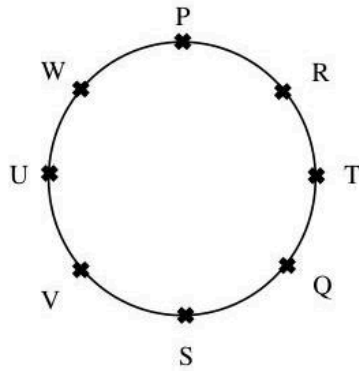
Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.7 Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1st round, 4th student behind P leaves the game. After 2nd round, 5th student behind Q leaves the game. After 3rd round, 3rd student behind V leaves the game. After 4th round, 4th student behind U leaves the game. Who all are left in the game after the 4th round?

Note: The figure shown is representative.



Options

- A. P; T; Q; S
- B. V; P; T; Q
- C. Q; T; V; W
- D. W; R; Q; V

Question Type : MCQ

Question ID : 1422761649

Status : Answered

Chosen Option : B

Q.8 Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II.

Group-I: Abuse → Insult → Ridicule

Group-II: _____ → Praise → Appreciate

Options

- A. Prize
- B. Extol
- C. Appropriate
- D. Espouse

Question Type : MCQ

Question ID : 1422761642

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.9 Had I learnt acting as a child, I _____ a famous film star.

Select the most appropriate option to complete the above sentence.

Options

- A. can be
- B. will be
- C. am going to be
- D. could have been

Question Type : MCQ

Question ID : 1422761643

Status : Answered

Chosen Option : A

Q.10 An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to ₹ 5,00,000. For higher fees, the overhead is ₹ 1,00,000 plus 10% of the amount by which the fee exceeds ₹ 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than ₹ 10,00,000, what is the maximum consulting fee that the employee can charge?

- Options
- A. ₹ 7,75,784
 - B. ₹ 7,51,232
 - C. ₹ 7,24,961
 - D. ₹ 7,01,438

Question Type : MCQ

Question ID : 1422761651

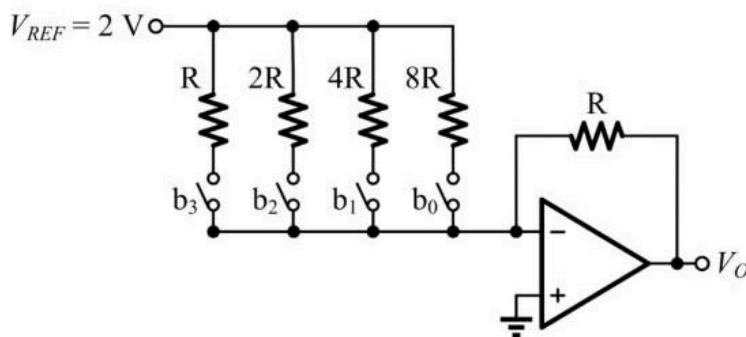
Status : Not Attempted and Marked For Review

Chosen Option : --

Section : EC Electronics and Communication Engineering

Q.1 A 4-bit weighted-resistor DAC with inputs b_3 , b_2 , b_1 , and b_0 (MSB to LSB) is designed using an ideal opamp, as shown below. The switches are closed when the corresponding input bits are logic '1' and open otherwise.

When the input $b_3b_2b_1b_0$ changes from 1110 to 1101, the magnitude of the change in the output voltage V_O (in mV, rounded off to the nearest integer) is _____.



Give 0.5
n
Ans
wer :

Question Type : NAT

Question ID : 1422761676

Status : Answered

Q.2 Consider a frequency-modulated (FM) signal

$$f(t) = A_c \cos(2\pi f_c t + 3 \sin(2\pi f_1 t) + 4 \sin(6\pi f_1 t)),$$

where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that $f_c > 100f_1$.

The peak frequency deviation of the FM signal in Hz is _____.

Options

- A. $2f_1$
- B. $15f_1$
- C. $12f_1$
- D. $4f_1$

Question Type : MCQ

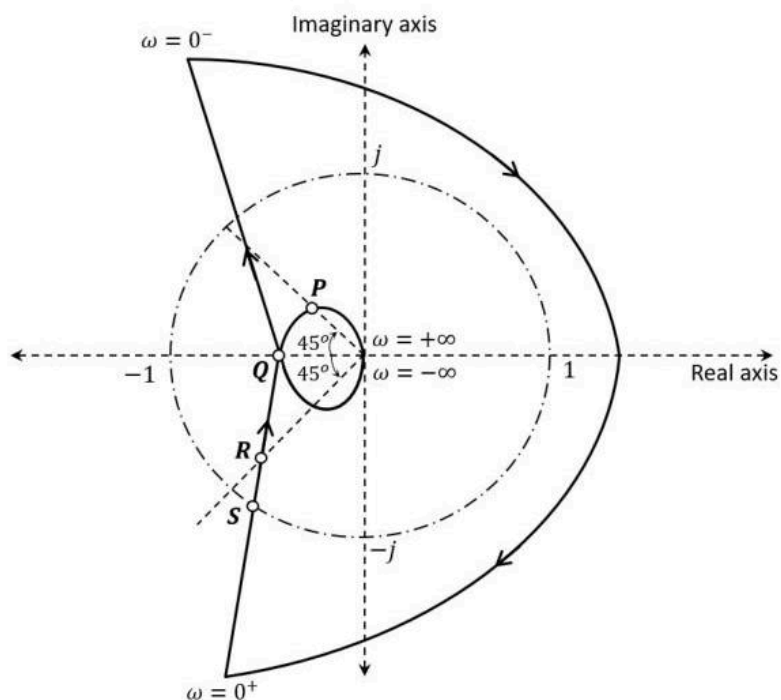
Question ID : 1422761655

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.3 The Nyquist plot of a system is given in the figure below. Let ω_P , ω_Q , ω_R , and ω_S be the positive frequencies at the points P , Q , R , and S , respectively.

Which one of the following statements is TRUE?



Options A.

ω_Q is the gain crossover frequency and ω_S is the phase crossover frequency

B.

ω_Q is the gain crossover frequency and ω_R is the phase crossover frequency

C.

ω_S is the gain crossover frequency and ω_Q is the phase crossover frequency

D.

ω_S is the gain crossover frequency and ω_P is the phase crossover frequency

Question Type : MCQ

Question ID : 1422761657

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.4

Let $x[n]$ be a discrete-time signal whose Z-transform is $X(z)$.

Which of the following statements is/are TRUE?

Options A.

If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the region of convergence (RoC) is the entire Z-plane

B.

The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle

C.

The discrete-time Fourier transform (DTFT) of $x[n]$ always exists

D.

The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros

Question Type : MSQ

Question ID : 1422761668

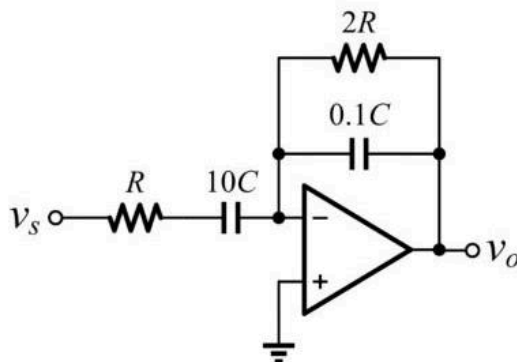
Status : Answered

Chosen Option : A,B

Q.5

All the components in the bandpass filter given below are ideal. The lower -3 dB frequency of the filter is 1 MHz.

The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____.



Give --
n
Ans
wer :

Question Type : NAT

Question ID : 1422761675

Status : Not Answered

Q.6 Which of the following statements is/are TRUE with respect to an ideal opamp?

Options

- A. It has an infinite output resistance
- B. It has an infinite open-loop common-mode gain
- C. It has an infinite input resistance
- D. It has an infinite open-loop differential gain

Question Type : MSQ

Question ID : 1422761670

Status : Answered

Chosen Option : A,D

Q.7 The generator matrix of a (6,3) binary linear block code is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

The minimum Hamming distance d_{min} between codewords equals ____ (answer in integer).

Give 1.

n
Ans
wer :

Question Type : NAT

Question ID : 1422761674

Status : Answered

Q.8

Consider the matrix A below:

$$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{bmatrix}$$

For which of the following combinations of α , β , and γ , is the rank of A at least three?

- (i) $\alpha = 0$ and $\beta = \gamma \neq 0$.
- (ii) $\alpha = \beta = \gamma = 0$.
- (iii) $\beta = \gamma = 0$ and $\alpha \neq 0$.
- (iv) $\alpha = \beta = \gamma \neq 0$.

Options

- A. Only (i) and (iii)
- B. Only (i), (iii), and (iv)
- C. Only (ii)
- D. Only (iv)

Question Type : MCQ

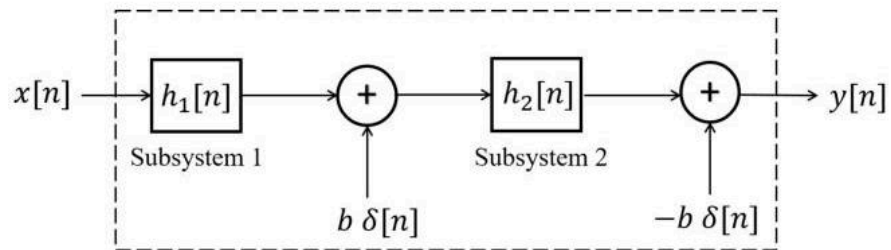
Question ID : 1422761652

Status : Answered

Chosen Option : A

Q.9 Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $b > 0$.

Assuming $h_2[n] \neq \delta[n]$, the overall system (denoted by the dashed box) is _____.



- Options
- A. nonlinear and time invariant
 - B. nonlinear and time variant
 - C. linear and time variant
 - D. linear and time invariant

Question Type : MCQ

Question ID : 1422761658

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.10 Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K \geq 0$. The root locus of this system is shown in Figure (ii) below.

For what value(s) of K will the system in Figure (i) have a pole at $-1 + j1$?

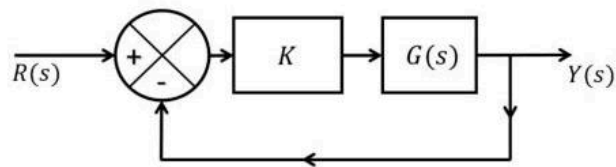


Figure (i)

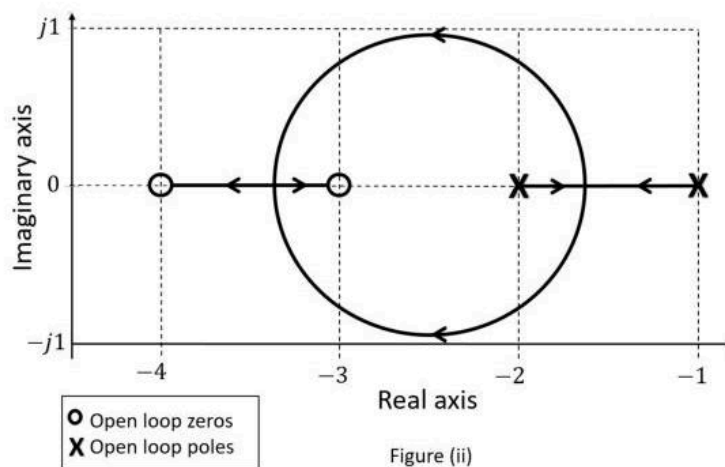


Figure (ii)

Options

- A. $K = 5$
- B. For all positive values of K
- C. $K = 1/5$
- D. For no positive value of K

Question Type : MSQ

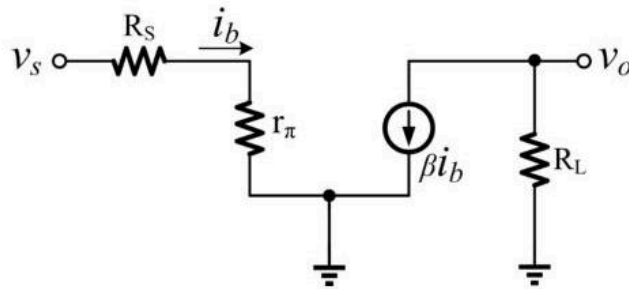
Question ID : 1422761667

Status : Answered

Chosen Option : A,D

Q.11 A simplified small-signal equivalent circuit of a BJT-based amplifier is given below.

The small-signal voltage gain V_o/V_s (in V/V) is _____.



Options

- A. $\frac{-\beta R_L}{R_S}$
- B. $\frac{-\beta R_L}{R_S + r_\pi}$
- C. $\frac{+\beta R_L}{R_S + r_\pi}$
- D. $\frac{+\beta R_L}{R_S}$

Question Type : MCQ

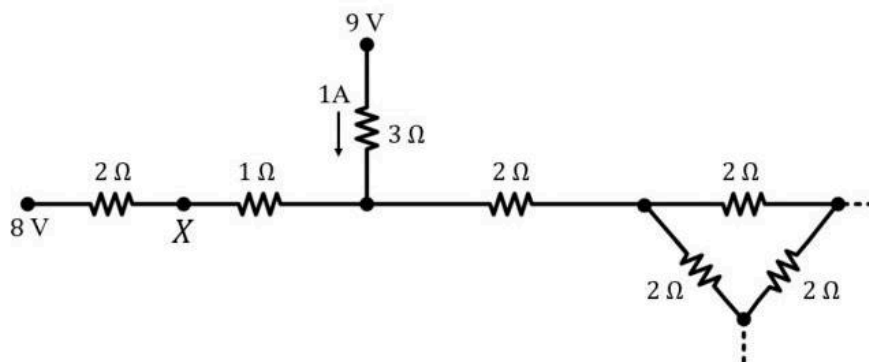
Question ID : 1422761662

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.12 Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3Ω resistor are as indicated.

The voltage (in Volts) at node X is _____.



Options

A. $\frac{20}{3}$

B. $\frac{32}{3}$

C. $\frac{22}{3}$

D. $\frac{2}{3}$

Question Type : MCQ

Question ID : 1422761660

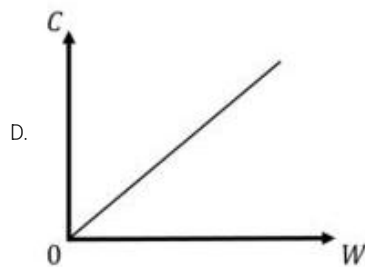
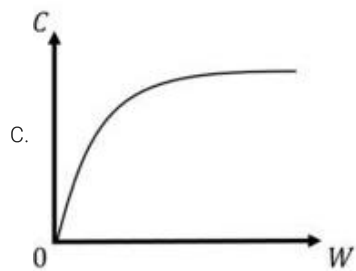
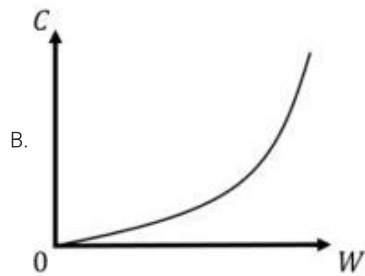
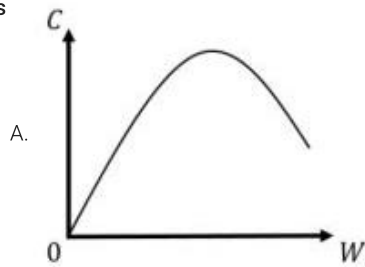
Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.13 Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\frac{N_0}{2}$. Let P_{av} denote the average transmit power constraint.

Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?

Options



Question Type : MCQ

Question ID : 1422761656

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.14 The function $y(t)$ satisfies

$$t^2 y''(t) - 2ty'(t) + 2y(t) = 0,$$

where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively.

Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0,1]$ is _____ (rounded off to two decimal places).

Give 11

n
Ans
wer :

Question Type : NAT

Question ID : 1422761673

Status : Marked For Review

Q.15 Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t) \exp(-j\omega t) dt$ exists.

Which one of the following statements is always TRUE?

Options

A. $|F(\omega)| \leq \int_{-\infty}^{\infty} |f(t)| dt$

B. $|F(\omega)| \leq \int_{-\infty}^{\infty} f(t) dt$

C. $|F(\omega)| \geq \int_{-\infty}^{\infty} f(t) dt$

D. $|F(\omega)| > \int_{-\infty}^{\infty} |f(t)| dt$

Question Type : MCQ

Question ID : 1422761659

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.16 A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement.

What is the probability that the two balls drawn have different colours?

Options

- A. 1
- B. $\frac{1}{3}$
- C. $\frac{1}{2}$
- D. $\frac{2}{3}$

Question Type : MCQ

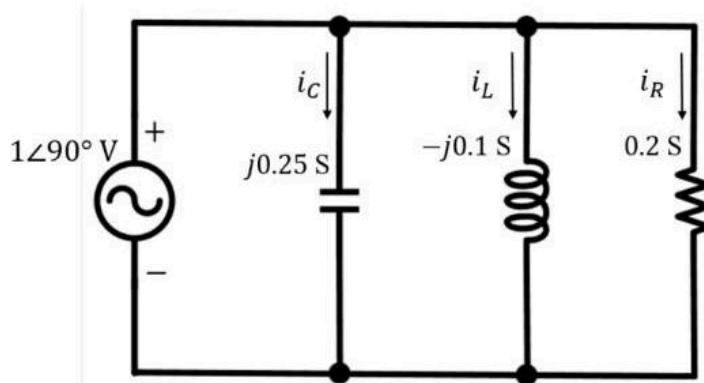
Question ID : 1422761654

Status : Answered

Chosen Option : A

Q.17 Let i_C , i_L , and i_R be the currents flowing through the capacitor, inductor, and resistor, respectively, in the circuit given below. The AC admittances are given in Siemens (S).

Which one of the following is TRUE?



Options

- A. $i_C = 0.25\angle 180^\circ \text{ A}$, $i_L = 0.1\angle 0^\circ \text{ A}$, $i_R = 0.2\angle 90^\circ \text{ A}$
- B. $i_C = 0.25\angle 270^\circ \text{ A}$, $i_L = 0.1\angle 90^\circ \text{ A}$, $i_R = 0.2\angle 90^\circ \text{ A}$
- C. $i_C = 4\angle 180^\circ \text{ A}$, $i_L = 10\angle 0^\circ \text{ A}$, $i_R = 5\angle 90^\circ \text{ A}$
- D. $i_C = 4\angle 90^\circ \text{ A}$, $i_L = 10\angle 270^\circ \text{ A}$, $i_R = 5\angle 0^\circ \text{ A}$

Question Type : MCQ

Question ID : 1422761661

Status : Marked For Review

Chosen Option : A

Q.18

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, defined as

$$f(x) = 2x^3 - 3x^2 - 12x + 1.$$

Which of the following statements is/are correct?

(Here, $\mathbb{R}$ is the set of real numbers.)

Options

- A. $x = -1$ is a local minimizer of f
- B. $x = 2$ is a local maximizer of f
- C. f has no global minimizer
- D. f has no global maximizer

Question Type : MSQ

Question ID : 1422761666

Status : Not Answered

Chosen Option : --

Q.19

Which of the following can be used as an n-type dopant for silicon?

Select the correct option(s).

Options

- A. Phosphorous
- B. Gallium
- C. Arsenic
- D. Boron

Question Type : MSQ

Question ID : 1422761672

Status : Answered

Chosen Option : B,C

Q.20

Consider the following series:

(i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$

Choose the correct option.

Options

- A. Only (iii) converges
- B. Only (ii) and (iii) converge
- C. All three converge
- D. Only (ii) converges

Question Type : MCQ

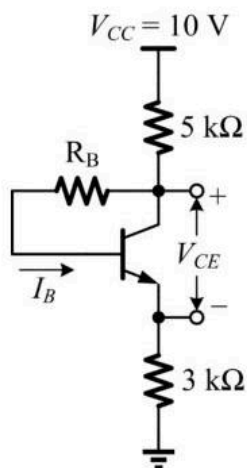
Question ID : 1422761653

Status : Answered

Chosen Option : A

Q.21 The ideal BJT in the circuit given below is biased in the active region with a β of 100.

If I_B is $10\ \mu\text{A}$, then V_{CE} (in Volts, rounded off to two decimal places) is _____.



Options

A. 4.95

B. 3.03

C. 3.73

D. 1.92

Question Type : MCQ

Question ID : 1422761663

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.22

Consider a message signal $m(t)$ which is bandlimited to $[-W, W]$, where W is in Hz. Consider the following two modulation schemes for the message signal:

- Double sideband-suppressed carrier (DSB-SC):

$$f_{DSB}(t) = A_c m(t) \cos(2\pi f_c t)$$

- Amplitude modulation (AM):

$$f_{AM}(t) = A_c (1 + \mu m(t)) \cos(2\pi f_c t)$$

Here, A_c and f_c are the amplitude and frequency (in Hz) of the carrier, respectively.

In the case of AM, μ denotes the modulation index.

Consider the following statements:

- (i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t .
- (ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t .

Which of the following options is/are correct?

Options

- A. (ii) is FALSE
- B. (i) is FALSE
- C. (i) is TRUE
- D. (ii) is TRUE

Question Type : MSQ

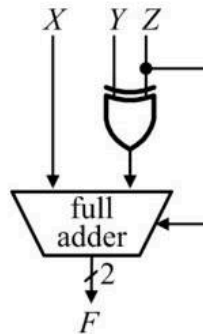
Question ID : 1422761669

Status : Answered

Chosen Option : A,B,C,D

Q.23 A full adder and an XOR gate are used to design a digital circuit with inputs X , Y , and Z , and output F , as shown below. The input Z is connected to the carry-in input of the full adder.

If the input Z is set to logic '1', then the circuit functions as _____ with X and Y as inputs.



- Options
- A. a binary to Gray code converter
 - B. an adder
 - C. a subtractor
 - D. a multiplier

Question Type : MCQ

Question ID : 1422761665

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.24 Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers having finite load resistors?

Options

- A. The common-source amplifier has an infinite input resistance
- B. The input and output voltages of the common-drain amplifier are in phase
- C. The common-gate amplifier has an infinite input resistance
- D. The input and output voltages of the common-source amplifier are in phase

Question Type : MSQ

Question ID : 1422761671

Status : Answered

Chosen Option : A,D

- Q.25 A 3-input majority logic gate has inputs X, Y , and Z . The output F of the gate is logic '1' if two or more of the inputs are logic '1'. The output F is logic '0' if two or more of the inputs are logic '0'.

Which one of the following options is a Boolean expression of the output F ?

Options

- A. $XY + YZ + ZX$
- B. $X \oplus Y \oplus Z$
- C. $X + Y + Z$
- D. XYZ

Question Type : MCQ

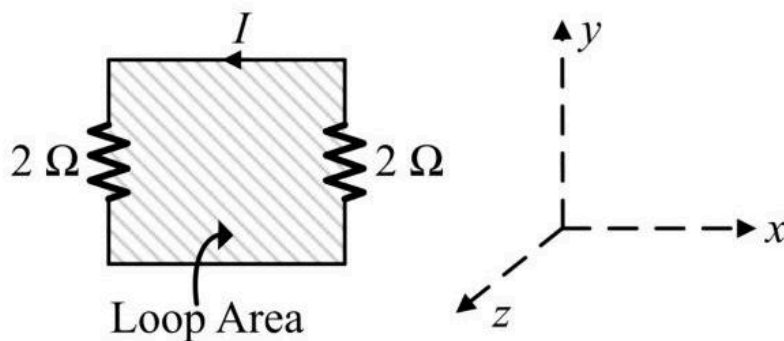
Question ID : 1422761664

Status : Not Attempted and Marked For Review

Chosen Option : --

- Q.26 Two resistors are connected in a circuit loop of area 5 m^2 , as shown in the figure below. The circuit loop is placed on the x - y plane.

When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z -axis, the magnitude of current I (in Amperes, rounded off to two decimal places) in the loop is _____.



Give 0.125

n
Ans
wer :

Question Type : NAT

Question ID : 1422761706

Status : Answered

Q.27 An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m.

The current (in Amperes, rounded off to two decimal places) flowing through the wire is _____.

Options

- A. 1.12
- B. 1.82
- C. 0.58
- D. 0.46

Question Type : MCQ

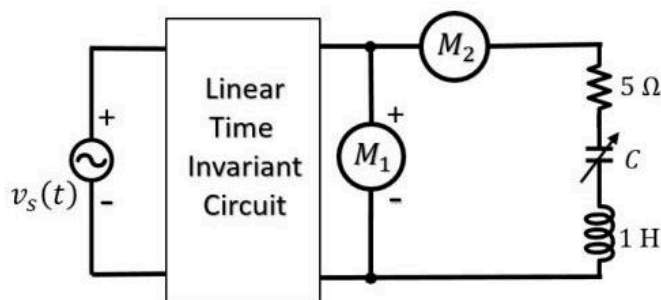
Question ID : 1422761691

Status : Not Attempted and Marked For Review

Chosen Option : --

Q.28 In the circuit below, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100 \cos(200t)$.

What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively?



Options

- A. $0.25\ \mu\text{F}$
- B. $25\ \mu\text{F}$
- C. Insufficient information to find C
- D. $4\ \mu\text{F}$

Question Type : MCQ

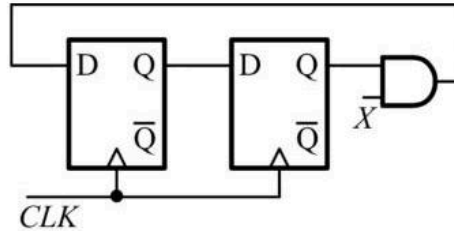
Question ID : 1422761680

Status : Not Attempted and Marked For Review

Chosen Option : --

Q.29 In the circuit shown below, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have a set-up time of 2 ns, a hold-time of 0 ns, and a clock-to-Q delay of 2 ns.

The maximum clock frequency (in MHz, *rounded off to the nearest integer*) such that there are no setup violations is _____.



Give 5
n
Ans
wer :

Question Type : NAT
Question ID : 1422761703
Status : Answered

Q.30 Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval.

Among the combinations of α and β given below, choose the one(s) for which the inequality

$$\beta \leq \int_2^8 f(x) dx \leq \alpha$$

is guaranteed to hold.

Options

- A. $\beta = 7m, \alpha = 6M$
- B. $\beta = 7m, \alpha = 5M$
- C. $\beta = 5m, \alpha = 7M$
- D. $\beta = 6m, \alpha = 5M$

Question Type : MSQ
Question ID : 1422761692
Status : Answered
Chosen Option : A,D

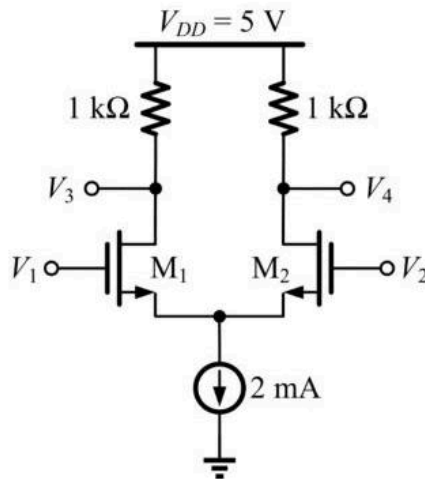
Q.31 The identical MOSFETs M_1 and M_2 in the circuit given below are ideal and biased in the saturation region. M_1 and M_2 have a transconductance g_m of 5 mS.

The input signals (in Volts) are:

$$V_1 = 2.5 + 0.01 \sin \omega t$$

$$V_2 = 2.5 - 0.01 \sin \omega t$$

The output signal V_3 (in Volts) is _____.



Options

- A. $3 + 0.05 \sin \omega t$
- B. $3 - 0.1 \sin \omega t$
- C. $4 + 0.1 \sin \omega t$
- D. $4 - 0.05 \sin \omega t$

Question Type : MCQ

Question ID : 1422761684

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.32 Consider a real-valued random process

$$f(t) = \sum_{n=1}^N a_n p(t - nT),$$

where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables.

Read the following statements about the random process and choose the correct option.

- (i) The mean of the process $f(t)$ is independent of time t .
- (ii) The autocorrelation function $E[f(t)f(t + \tau)]$ is independent of time t for all τ .

(Here, $E[\cdot]$ is the expectation operation.)

Options

- A. (i) is FALSE and (ii) is TRUE
- B. Both (i) and (ii) are FALSE
- C. Both (i) and (ii) are TRUE
- D. (i) is TRUE and (ii) is FALSE

Question Type : MCQ

Question ID : 1422761683

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.33 The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} / \text{m}^3$ at 300 K.

If the electron and hole mobilities are $0.15 \text{ m}^2/\text{Vs}$ and $0.05 \text{ m}^2/\text{Vs}$, respectively, then the intrinsic resistivity of the semiconductor (in $\text{k}\Omega\cdot\text{m}$) at 300 K is _____.

(Charge of an electron $e = 1.6 \times 10^{-19} \text{ C}$.)

Options

- A. 1.25
- B. 0.85
- C. 1.65
- D. 1.95

Question Type : MCQ

Question ID : 1422761687

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.34 X and Y are Bernoulli random variables taking values in $\{0,1\}$. The joint probability mass function of the random variables is given by:

$$\begin{aligned}P(X = 0, Y = 0) &= 0.06 \\P(X = 0, Y = 1) &= 0.14 \\P(X = 1, Y = 0) &= 0.24 \\P(X = 1, Y = 1) &= 0.56\end{aligned}$$

The mutual information $I(X; Y)$ is _____ (rounded off to two decimal places).

Give 1.13

n
Ans
wer :

Question Type : NAT

Question ID : 1422761701

Status : Answered

Q.35 An ideal p-n junction germanium diode has a reverse saturation current of $10 \mu\text{A}$ at 300 K.

The voltage (in Volts, rounded off to two decimal places) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____.

(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19} \text{C}$.)

Give --

n
Ans
wer :

Question Type : NAT

Question ID : 1422761704

Status : Not Answered

Q.36 Let $f(t)$ be a periodic signal with fundamental period $T_0 > 0$. Consider the signal $y(t) = f(\alpha t)$, where $\alpha > 1$.

The Fourier series expansions of $f(t)$ and $y(t)$ are given by

$$f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\frac{2\pi}{T_0}kt} \quad \text{and} \quad y(t) = \sum_{k=-\infty}^{\infty} d_k e^{j\frac{2\pi}{T_0}\alpha kt}.$$

Which of the following statements is/are TRUE?

Options

- A. $c_k = d_k/\alpha$ for all k
- B. $c_k = d_k$ for all k
- C. $y(t)$ is periodic with a fundamental period T_0/α
- D. $y(t)$ is periodic with a fundamental period αT_0

Question Type : MSQ

Question ID : 1422761696

Status : Answered

Chosen Option : B,D

Q.37 Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as follows.

L is the number of roots of $p(s)$ with negative real parts.

I is the number of roots of $p(s)$ that are purely imaginary.

R is the number of roots of $p(s)$ with positive real parts.

Which one of the following options is correct?

Options

- A. $L = 2$, $I = 2$, and $R = 1$
- B. $L = 1$, $I = 2$, and $R = 2$
- C. $L = 3$, $I = 2$, and $R = 0$
- D. $L = 0$, $I = 4$, and $R = 1$

Question Type : MCQ

Question ID : 1422761678

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.38 Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?

Options

- A. $\oint_C e^z dz = 0$
- B. $\oint_C \sec z dz \neq 0$
- C. $\oint_C \cos z dz \neq 0$
- D. $\oint_C z^n dz = 0$, where n is an even integer

Question Type : MSQ

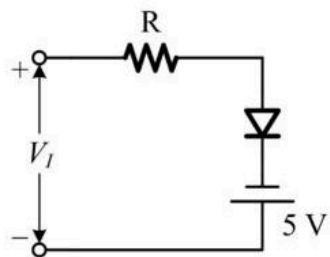
Question ID : 1422761693

Status : Answered

Chosen Option : A,C,D

Q.39 The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $V_I = 10 \sin 100\pi t$, where time t is in seconds.

The time duration (in ms, rounded off to two decimal places) for which the diode is forward biased during one period of the input is _____.



Give 0.87265

n
Ans
wer :

Question Type : NAT

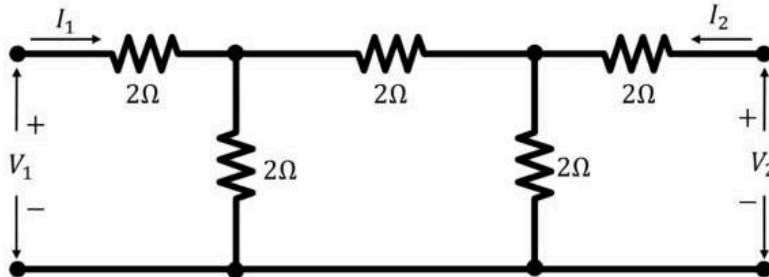
Question ID : 1422761702

Status : Answered

Q.40 The Z-parameter matrix of a two port network relates the port voltages and port currents as follows:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = Z \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

The Z-parameter matrix (with each entry in Ohms) of the network shown below is _____.



Options

A. $\begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$

B. $\begin{bmatrix} 10/3 & 1/3 \\ 1/3 & 10/3 \end{bmatrix}$

C. $\begin{bmatrix} 2/3 & 10/3 \\ 10/3 & 2/3 \end{bmatrix}$

D. $\begin{bmatrix} 10 & 2 \\ 2 & 10 \end{bmatrix}$

Question Type : MCQ

Question ID : 1422761681

Status : Not Attempted and Marked For Review

Chosen Option : --

Q.41 A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S . The receiver uses the maximum likelihood decoder to estimate the transmitted symbol $\hat{S}$.

Suppose the probability of symbol estimation error P_e is expressed as follows:

$$P_e = \alpha P(N > 1),$$

where $P(N > 1)$ denotes the probability that N exceeds 1.

What is the value of α ?

Options

- A. 1
- B. $\frac{2}{3}$
- C. $\frac{1}{3}$
- D. $\frac{4}{3}$

Question Type : MCQ

Question ID : 1422761682

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.42 A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V.

For an input sinusoidal signal with frequency 500 kHz, the maximum SNR (in dB, rounded off to two decimal places) and the data rate (in Mbps) at the output of the ADC are _____, respectively.

Options

- A. 61.96 and 10
- B. 33.36 and 5
- C. 61.96 and 5
- D. 33.36 and 10

Question Type : MCQ

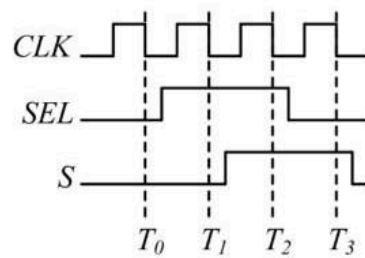
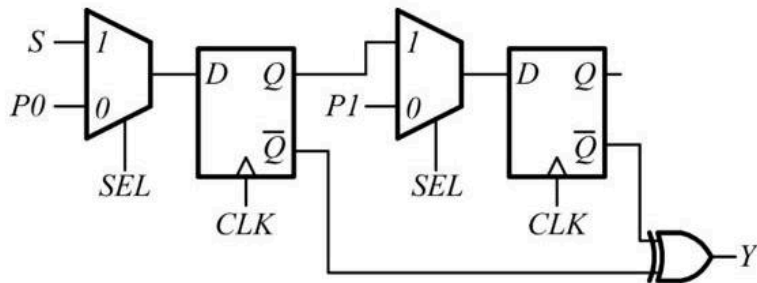
Question ID : 1422761685

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.43 A positive-edge-triggered sequential circuit is shown below. There are no timing violations in the circuit. Input $P0$ is set to logic '0' and $P1$ is set to logic '1' at all times. The timing diagram of the inputs SEL and S are also shown below.

The sequence of output Y from time T_0 to T_3 is _____.



Options

- A. 1011
- B. 1101
- C. 0100
- D. 0010

Question Type : MCQ

Question ID : 1422761686

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.44 Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec.

The signal $f(t)$ is uniformly sampled to obtain $y(t) = f(t)p(t)$. Here,

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s),$$

with $\delta(t)$ being the Dirac impulse, $T_s > 0$, and $\tau > 0$. The sampled signal $y(t)$ is passed through an ideal lowpass filter $h(t) = \omega_c T_s \frac{\sin(\omega_c t)}{\pi \omega_c t}$ with cutoff frequency ω_c and passband gain T_s .

The output of the filter is given by _____.

Options

- A. $f(t - \tau)$ if $T_s < 2\pi/\omega_c$
- B. $f(t - \tau)$ if $T_s < \pi/\omega_c$
- C. $T_s f(t)$ if $T_s < 2\pi/\omega_c$
- D. $f(t)$ if $T_s < \pi/\omega_c$

Question Type : MCQ

Question ID : 1422761679

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.45

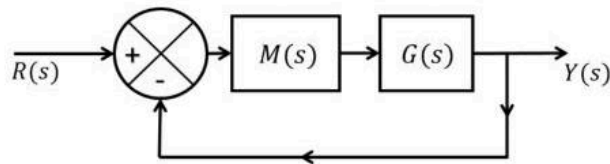
Let $G(s) = \frac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to the system $G(s)$ in the configuration shown below.

Consider the following statements.

(i) There exists no controller of the form $M(s) = \frac{K_I}{s}$, where K_I is a positive real number, such that the closed loop system is stable.

(ii) There exists at least one controller of the form $M(s) = K_P + sK_D$, where K_P and K_D are positive real numbers, such that the closed loop system is stable.

Which one of the following options is correct?



Options

- A. Both (i) and (ii) are TRUE
- B. (i) is TRUE and (ii) is FALSE
- C. (i) is FALSE and (ii) is TRUE
- D. Both (i) and (ii) are FALSE

Question Type : MCQ

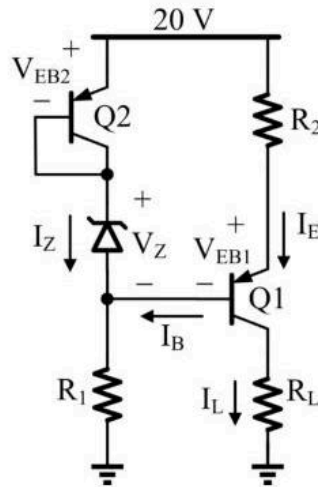
Question ID : 1422761677

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.46 In the circuit shown, the identical transistors Q1 and Q2 are biased in the active region with $\beta = 120$. The Zener diode is in the breakdown region with $V_Z = 5\text{ V}$ and $I_Z = 25\text{ mA}$.

If $I_L = 12\text{ mA}$ and $V_{EB1} = V_{EB2} = 0.7\text{ V}$, then the values of R_1 and R_2 (in $k\Omega$, rounded off to one decimal place) are _____, respectively.



- Options
- A. 0.6 and 0.4
 - B. 6.0 and 4.0
 - C. 1.4 and 2.5
 - D. 14.0 and 25.0

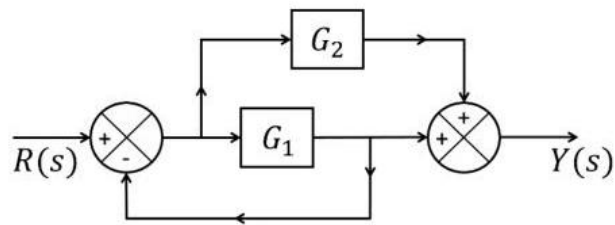
Question Type : MCQ

Question ID : 1422761688

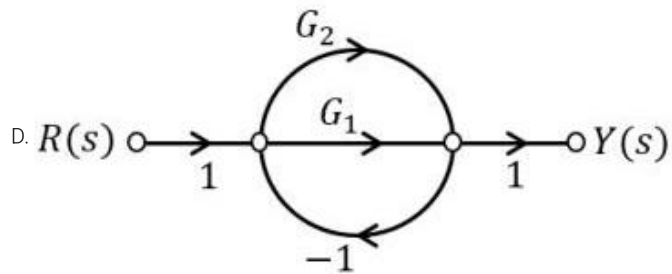
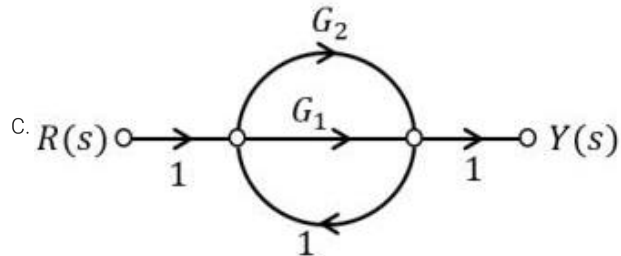
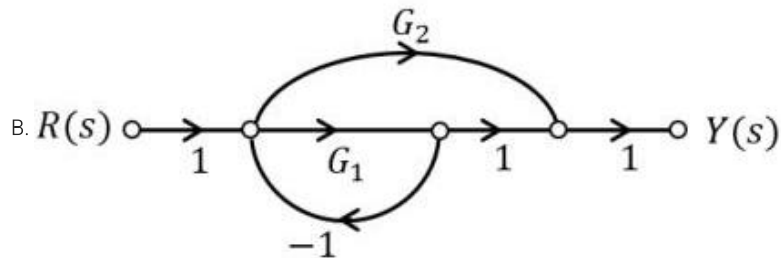
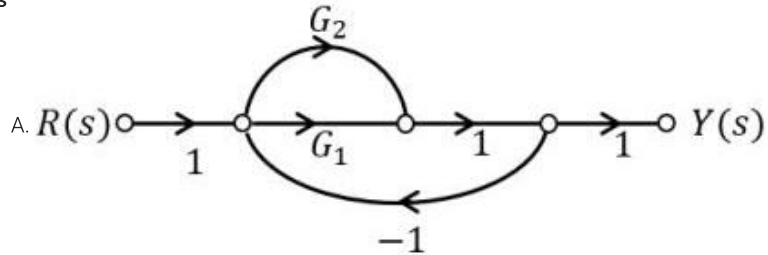
Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.47 Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).



Options



Question Type : MSQ

Question ID : 1422761697

Status : Answered

Chosen Option : B,D

Q.48 Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained.

The expectation of X is _____ (rounded off to two decimal places).

Give 9
n
Ans
wer :

Question Type : NAT
Question ID : 1422761699
Status : Answered

Q.49 The random variable X takes values in $\{-1, 0, 1\}$ with probabilities

$$P(X = -1) = P(X = 1) = \alpha \text{ and } P(X = 0) = 1 - 2\alpha, \text{ where } 0 < \alpha < \frac{1}{2}.$$

Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α .

Which of the following statements is/are TRUE?

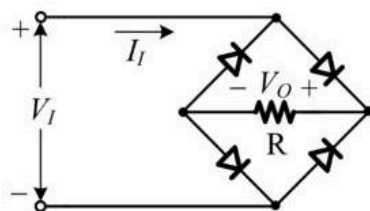
Options

- A. $g(0.3) > g(0.4)$
- B. $g(0.3) > g(0.25)$
- C. $g(0.25) > g(0.3)$
- D. $g(0.4) > g(0.3)$

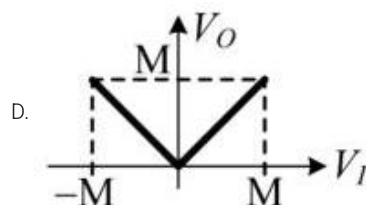
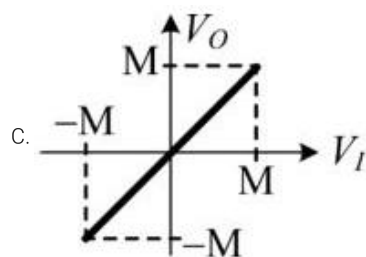
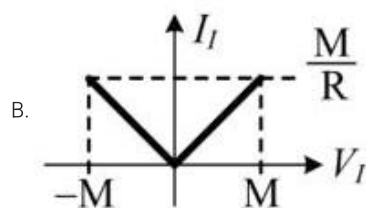
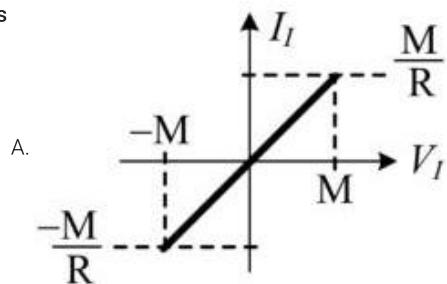
Question Type : MSQ
Question ID : 1422761695
Status : Answered
Chosen Option : B,D

Q.50 All the diodes in the circuit given below are ideal.

Which of the following plots is/are correct when V_I (in Volts) is swept from $-M$ to M ?



Options



Question Type : MSQ

Question ID : 1422761698

Status : Answered

Chosen Option : A,B,D

Q.51 Consider the vectors

$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 3\sqrt{2} \end{bmatrix}.$$

For real-valued scalar variable x , the value of

$$\min_x \|\mathbf{a}x - \mathbf{b}\|_2$$

is _____ (rounded off to two decimal places).

$\|\cdot\|_2$ denotes the Euclidean norm, i.e., for $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, $\|\mathbf{y}\|_2 = \sqrt{y_1^2 + y_2^2}$.

Give 15

n
Ans
wer :

Question Type : NAT

Question ID : 1422761700

Status : Answered

Q.52 Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential equations governing the system are given by

$$\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t).$$

Which of the following statements is/are TRUE?

Options A.

The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded for all bounded inputs

B.

There exists a bounded input such that the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded

C.

There exists a bounded input such that at least one of the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ is unbounded

D.

The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs

Question Type : MSQ

Question ID : 1422761694

Status : Answered

Chosen Option : B,C,D

Q.53 The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is $0.38 \text{ m}^2/\text{Vs}$.

The electron diffusivity D_n at 300 K (in cm^2/s , rounded off to the nearest integer) is _____.

(Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{ J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19} \text{ C}$.)

Options

A. 98

B. 38

C. 10

D. 26

Question Type : MCQ

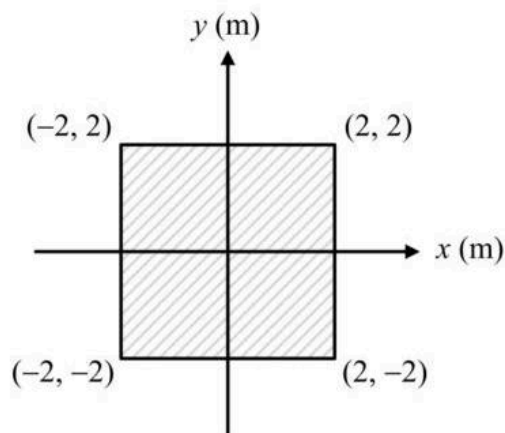
Question ID : 1422761689

Status : Not Attempted and
Marked For Review

Chosen Option : --

Q.54 A square metal sheet of $4\text{ m} \times 4\text{ m}$ is placed on the x - y plane as shown in the figure below.

If the surface charge density (in $\mu\text{C}/\text{m}^2$) on the sheet is $\rho_s(x, y) = 4|y|$, then the total charge (in μC , rounded off to the nearest integer) on the sheet is _____.



Options A. 256

B. 64

C. 85

D. 16

Question Type : MCQ

Question ID : 1422761690

Status : Answered

Chosen Option : B

Q.55 A $50\ \Omega$ lossless transmission line is terminated with a load Z_L of $(50 - j75)\ \Omega$.

If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, rounded off to one decimal place) is _____.

Give 10

n

Ans

wer :

Question Type : NAT

Question ID : 1422761705

Status : Answered